Absolute Precision Is Not Enough: Bayesian Stopping Under Rare Relevance Prevalence

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Abstract

High-recall retrieval tasks such as Technology-Assisted Review and systematic-review screening often operate under severe class imbalance, where relevant documents form only a small fraction of the collection. When relevance judgments are costly, assessment may be stopped once uncertainty about relevance prevalence is considered sufficiently small. We study the limitations of stopping based on a fixed absolute width of a posterior credible interval. Using a Beta–Bernoulli model with Highest Posterior Density intervals, we simulate sequential assessment over prevalence values from 0.5 to 10^{-4}. Results show a systematic mismatch between absolute posterior precision and relative estimation accuracy: under extreme imbalance, assessment may terminate after very few judgments, sometimes before any relevant document is observed, while relative error remains large. The findings show that stopping criteria should be aligned with the scale on which estimation accuracy matters, with direct implications for prevalence estimation and assessment stopping in high-recall IR. The source code of the experiments can be found here https://anonymous.4open.science/r/IIR2026-FEE8/.

Keywords

Technology-Assisted Review, high-recall retrieval, relevance prevalence, Bayesian estimation, adaptive stopping, class imbalance

1. Introduction

High-recall information retrieval (IR) aims to identify as many relevant documents as possible while controlling the human effort required to assess them. This setting appears in Technology-Assisted Review (TAR), systematic reviews, legal discovery, patent search, and other screening tasks in which relevance judgments are expensive and exhaustive assessment is generally infeasible [1, 2, 3]. A central question is therefore when to stop assessing documents.

Most TAR stopping methods seek to determine whether a recall target has been reached or whether further review is justified by its cost. Their operational criteria differ: examples include statistical certification, counting-process models, heuristic rules, and decision-theoretic approaches [4, 5, 3, 6]. Some methods estimate the total or residual number of relevant documents to support recall estimation [2, 7]. None of these categories should be conflated with the simple absolute posterior-width rule investigated here. Nevertheless, prevalence is an important latent quantity in high-recall retrieval, even when it is not the final quantity of interest.

We focus here on a narrower question: when has relevance prevalence itself been estimated precisely enough? Let $p = \Pr(Y = 1)$ denote the proportion of relevant documents in a population. The meaning of a fixed precision requirement changes in a remarkable way with prevalence. An absolute tolerance of 0.01, for example, corresponds to only 2% of $p = 0.5$, but to 100 times a prevalence of 10^{-4}. This raises a potential problem for stopping criteria defined only through absolute interval width.

Bayesian models have long been used in IR to quantify uncertainty when exhaustive relevance assessment is unavailable; for example, Webber used beta-binomial posteriors to construct recall intervals from sampled judgments [8]. More recently, HPD credible intervals have been used in data-quality assessment to stop annotation once an absolute margin-of-error constraint is satisfied [9]. The distinction between absolute and relative precision is established in statistical estimation; we do not claim it as a new statistical result or propose a new Bayesian estimator. Our contribution to IR is a controlled characterization of the practical consequences of applying an absolute-width assessment

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rule across realistic orders of magnitude of relevance prevalence. We jointly examine review effort, relative estimation error, interval coverage, and the number of observed relevant documents. The Beta–Bernoulli model serves as a diagnostic instrument for this purpose, not as a model of a complete TAR workflow.

We address three research questions:

- **RQ1:** How does relevance prevalence affect the assessment effort required by absolute posterior-width stopping?
- **RQ2:** Does satisfying an absolute posterior-width criterion imply accurate prevalence estimation under extreme relevance imbalance?
- **RQ3:** How much positive relevance evidence has been accumulated when stopping occurs?

Our main finding is that absolute posterior precision can be poorly aligned with proportional estimation accuracy. As relevance prevalence decreases, an interval can satisfy the same absolute-width requirement after observing very little evidence about the relevant class, while relative estimation error remains large.

2. Background and Problem Formulation

2.1. Stopping in high-recall retrieval

Stopping has been studied extensively in TAR. Li and Kanoulas jointly rank documents and sample from an adaptive distribution to estimate the residual number of relevant documents [1]. Related work has also shown how prevalence estimation can be combined with ranking information to estimate recall early and derive conservative stopping rules under limited assessment budgets [10]. Other approaches estimate the total number of relevant documents or certify that a target recall has been reached using statistical evidence from sampled judgments [7, 4, 5]. A further line of work models the rate at which relevant documents are encountered using counting or point processes [3, 2]. More recent approaches question whether recall alone should determine termination, and instead use confidence about downstream conclusions or decision-theoretic utility [11, 6]. These methods differ in their statistical machinery and operational objective, but all illustrate that stopping should be tied to the information requirement of the task.

Our problem is complementary. We do not seek to certify that a target recall has been reached, nor do we assert that the fixed absolute-width rule is standard practice in TAR. The direct methodological precedent is posterior-width stopping in related annotation and data-quality assessment [9]; TAR provides the motivating setting in which prevalence estimation may inform downstream decisions. We study assessment used to estimate the base rate of relevance itself. This quantity can be extremely small in high-recall settings and directly affects how much evidence can be expected from random assessment. Under the Bernoulli sampling model considered below, after n assessments the expected number of relevant observations is $\mathbb{E}[S_n] = np$.

For $p = 10^{-4}$, even 1,000 assessments yield only 0.1 relevant documents in expectation. A stopping criterion that can terminate much earlier may therefore report small absolute uncertainty without having accumulated meaningful evidence about the scale of the rare prevalence.

2.2. Bayesian prevalence estimation and absolute-width stopping

Let $Y_i \in \{0, 1\}$ indicate whether the ith assessed document is relevant, with $Y_i \sim \mathrm{Bernoulli}(p)$. We adopt a Beta prior $p \sim \mathrm{Beta}(\alpha, \beta)$. After observing s relevant and f non-relevant documents, conjugacy gives $p \mid s, f \sim \mathrm{Beta}(\alpha + s, \beta + f)$. The posterior mean is

$$\hat{p} = \mathbb{E}[p \mid s, f] = \frac{\alpha + s}{\alpha + \beta + s + f}.$$

At each step we compute a 95% Highest Posterior Density (HPD) credible interval $[H_L, H_U]$ and its width

$$w = H_U - H_L.$$

Assessment stops as soon as

$$w \leq \tau, \tag{1}$$

where τ is an absolute uncertainty tolerance on the probability scale.

HPD intervals provide a compact summary of posterior uncertainty, and absolute-width stopping has been used in related assessment settings [9]. The rule in Eq. (1), however, is defined on the absolute probability scale: the same value of τ is applied regardless of whether p is close to 0.5 or several orders of magnitude smaller. Our aim is to examine the operating behaviour of this criterion as p approaches zero.

3. Experimental Design

We isolate prevalence through controlled simulation. For each true prevalence $p \in \{0.5, 0.1, 0.01, 0.001, 0.0001\}$, we generate sequential Bernoulli observations. We use an uninformative $\mathrm{Beta}(1, 1)$ prior, a 95% HPD interval, and thresholds $\tau \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$. Each configuration is repeated $1{,}000$ times. A run is terminated if it has not met the width criterion after $10{,}000$ assessments; such runs are recorded as budget-limited.

We report four quantities:

- Assessment effort is the stopping sample size n_{stop}.
- To check whether the final posterior intervals still behave sensibly across repeated simulations, we report empirical coverage $\mathbb{I}(H_L \leq p \leq H_U)$.
- The central accuracy measure for the present study is relative error, $RE = \frac{|\hat{p}-p|}{p}$, because an identical absolute error has radically different meaning at $p = 0.5$ and $p = 10^{-4}$.
- Finally, we record the number of relevant observations $S_{n_{\mathrm{stop}}}$ accumulated before termination. This last quantity is especially informative in rare-event regimes.

Simulation is deliberate in this diagnostic study: it lets us vary only the true relevance prevalence over several orders of magnitude while keeping the inference procedure and assessment budget fixed. The Bernoulli assumption corresponds to independent judgments with a constant relevance probability, as in simple random sampling with replacement. It does not represent ranked or adaptively selected documents, heterogeneous topic distributions, or classifier-assisted review. Consequently, these experiments diagnose a property of the specified stopping rule; they do not measure the reliability of existing TAR stopping systems. Real IR collections will be needed to study interactions with ranking, topic heterogeneity, finite populations, and non-random assessment.

4. Results

4.1. RQ1: Assessment effort can decrease as the task becomes more imbalanced

Figure 1 summarizes the stopping behaviour across prevalence levels. The left panel shows the mean number of assessments at termination, either because the absolute-width criterion is satisfied or because the budget of $10{,}000$ assessments is exhausted. Stricter values of τ generally require more evidence, as expected. More importantly, for a fixed absolute-width threshold the procedure can terminate substantially earlier as prevalence approaches zero. With $\tau = 0.1$, the mean number of assessments falls from approximately 380 at $p = 0.5$ to only 28 at $p = 10^{-4}$. Even with $\tau = 0.01$, the corresponding rare-prevalence condition terminates after roughly 302 assessments.

This behaviour is consistent with the scale of Bernoulli uncertainty, whose variance is governed by $p(1 - p)$. Near the boundary, uncertainty measured on the absolute probability scale can become

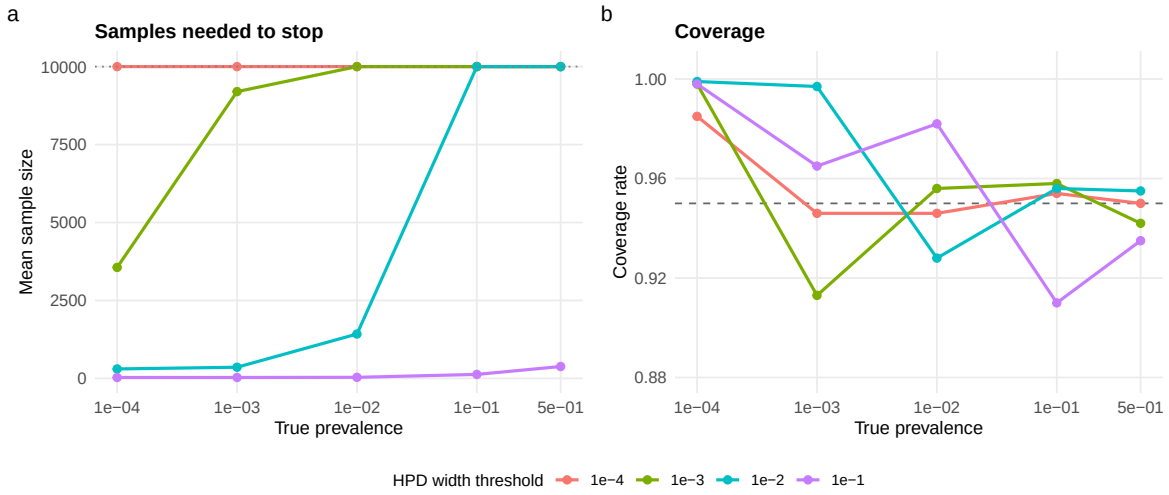


Figure 1: Stopping behaviour across prevalence levels. Left: mean number of assessments at termination for each absolute HPD-width threshold; values at the $10{,}000$-assessment ceiling indicate budget-limited runs that may not have satisfied the stopping condition. Right: empirical coverage of the final 95% HPD intervals; the dashed line denotes the nominal 0.95 level.

small with comparatively few observations. Consequently, an absolute-width criterion can make rare-prevalence estimation appear "easy", even though determining the magnitude of p with useful relative accuracy requires substantially more evidence about the rare relevant class.

The right panel of Figure 1 reports empirical coverage of the final 95% HPD intervals. Coverage remains broadly close to the nominal level, although it is not uniformly calibrated across configurations and ranges approximately from 0.91 to 1.00. Importantly, the large changes in assessment effort are not accompanied by a comparable collapse in coverage. This indicates that the main issue is not simply whether the final interval contains the true prevalence, but whether the stopping criterion measures precision on an appropriate scale.

4.2. RQ2: Absolute precision does not imply relative accuracy

The apparent efficiency in Figure 1 masks a sharp deterioration in relative estimation accuracy. The left panel of Figure 2 shows that mean relative error increases by orders of magnitude as prevalence decreases. For $\tau = 0.1$, it is approximately 2.6 at $p = 0.01$, 32.7 at $p = 0.001$, and 332.5 at $p = 10^{-4}$. Even the substantially stricter $\tau = 0.001$ criterion yields a mean relative error of approximately 2.7 at $p = 10^{-4}$.

This is the central failure mode examined in this study. The stopping rule is doing what it is designed to do: terminate once the posterior interval has an absolute width no larger than τ. However, the criterion is not scale invariant. The same absolute tolerance that provides meaningful precision for moderate prevalence can be practically uninformative when the relevant class is several orders of magnitude rarer.

4.3. RQ3: Stopping may occur before relevant evidence is observed

The right panel of Figure 2 helps explain the large relative errors observed under extreme imbalance. At moderate prevalence, many relevant documents are observed before termination. In rare-relevance regimes, however, the absolute-width criterion can be satisfied with essentially no evidence from the relevant class. With $p = 10^{-4}$ and $\tau = 0.1$, the procedure terminates after about 28 assessments and observes, on average, essentially no relevant documents. At $\tau = 0.01$, termination occurs after roughly 302 assessments while still yielding far fewer than one relevant document on average.

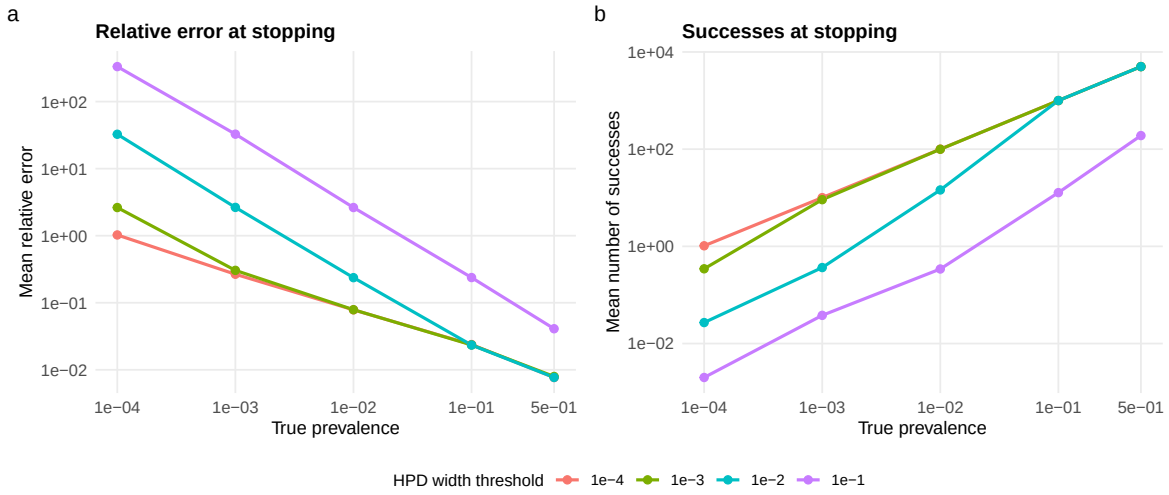


Figure 2: Estimation behaviour at termination. Left: mean relative prevalence-estimation error. Right: mean number of relevant documents observed by termination. Both panels use logarithmic scaling to show variation across prevalence regimes.

The posterior has therefore accumulated evidence that p lies near the lower boundary of the probability scale, but not enough evidence to determine its magnitude with useful relative accuracy. This distinction is particularly important in high-recall retrieval: establishing that relevant documents are rare is not equivalent to distinguishing, for example, a prevalence of 10^{-2} from 10^{-3} or 10^{-4}.

Taken together, the results expose a systematic tension: as prevalence becomes rarer, absolute-width stopping may require fewer assessments precisely in the regime where relative estimation becomes harder and less relevant-document evidence is available to support the estimate.

5. Implications for High-Recall Retrieval

The results suggest that assessment stopping should not be defined independently of the scale on which estimation error matters. Importantly, they establish a failure mode for the *evaluated absolute-width criterion*, not a general failure of confidence-based or Bayesian stopping in TAR. Operational TAR methods often target recall, residual relevance, or decision utility rather than the width of a prevalence interval alone. In TAR and related high-recall tasks, relevance prevalence may be a nuisance quantity used to estimate the total number of relevant documents, residual relevant mass, or recall. In other settings, prevalence itself may be of interest, for example when characterizing topic-level relevance prevalence or allocating assessment effort across topics. In either case, an estimate that is precise only in absolute terms may be insufficient when relevant documents are rare.

We emphasize that this paper does not establish that relative-width stopping is universally preferable. Relative criteria themselves become unstable near zero, while inverse-sampling, evidence-based, confidence-sequence, and decision-theoretic alternatives embody different guarantees and operational objectives. Classical sequential statistics has long distinguished absolute from relative precision; our results show why this distinction deserves explicit attention in IR assessment. The natural next step is therefore to compare stopping objectives and their guarantees, rather than merely alternative interval estimators.

The implications also extend to budget allocation. If topics differ substantially in relevance prevalence, then a common absolute precision threshold can impose very different effective accuracy requirements. A topic with $p = 0.1$ and one with $p = 10^{-4}$ should not automatically be considered equally well estimated merely because their posterior intervals have the same width. More generally, the results suggest that stopping thresholds and assessment budgets should be aligned with topic-specific inferential goals, complementing recent work on budget-aware TAR [6, 12, 13].

6. Conclusions

We studied a simple Bayesian sequential assessment procedure to diagnose the behaviour of absolute posterior-width stopping under rare relevance prevalence. The experiments reveal a clear tension: as prevalence decreases, the absolute-width stopping criterion can be satisfied with comparatively few assessments, while relative estimation error increases dramatically and few or no relevant documents may have been observed. The result is not a failure of Bayesian inference, but of objective alignment. Absolute precision is appropriate when absolute error is the quantity of interest; it can be poorly aligned with tasks that require proportional accuracy for a rare class. For high-recall IR, where relevance may be extremely sparse and assessment budgets are limited, stopping criteria should therefore make the intended notion of estimation quality explicit. Future work will compare absolute, relative, evidence-based, and sequentially valid stopping criteria on TREC, CLEF, and TAR collections. A key question will be whether the mismatch observed here persists when prevalence estimates are coupled to ranking, adaptive assessment, and recall-based stopping decisions.

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Declaration on Generative AI

During the preparation of this work, the author(s) used GPT-6 in order to: Grammar and spelling check. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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